

Jinpeng Lu

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Video World Models | Generative-Video Evaluation | Embodied AI



📌 Research Profile & Core Strengths

My research focuses on **video world models** and **generative-video evaluation** and **embodied intelligence** / **vision-language-action models**, with earlier work in medical AI. I also have an exploratory interest in the verifiability, safety, and evaluation of **recursive self-improvement (RSI)**.

My work turns open-ended capability questions into reproducible diagnostics, including WRBench world-state evaluation and collaborative research on VLA representation / behavior tracing, alongside VeloxSeg and H2ASeg for efficient 3D perception.

Experienced with PyTorch / DDP / CUDA / Linux training and evaluation pipelines, data construction, custom model implementation, ablation analysis, visualization, and English academic manuscript writing.

🎓 Education

University of Science and Technology of China

Sep. 2025 – Present

M.S. in Information and Communication Engineering, expected Jun. 2028

Advisor: Zhiwei Xiong; interests: video world models, generative-video evaluation, embodied AI / VLA models; exploratory interest in verifiability, safety, and evaluation of RSI.

Harbin Institute of Technology, Shenzhen

Sep. 2021 – Jun. 2025

B.S. in Data Science and Big Data Technology; GPA: 3.728/4, Rank: 17/172, Top 10%

Core training: mathematics, statistics, numerical computing, data engineering; research training: PET/CT 3D segmentation, generative signal enhancement.

🏆 Representative Works

Video World Models & Generative-Video Evaluation

Current World Models Lack a Persistent State Core

arXiv preprint, First Author

2026.06

- **Task:** Test whether video world models keep event-consistent state across visible, hidden, and re-observed phases under viewpoint changes.
- **Method:** Built **WRBench** with camera motion as an **observability intervention**, separating camera execution, visible consistency, re-observation support, and returned-state correctness.
- **Result:** Evaluated **23 models** / **9,600 videos** with **2,547 human verdicts**; clear frames and camera control still failed to ensure endpoint binding.

Embodied Intelligence & VLA

VLA-Trace: Diagnosing Vision-Language-Action Models through Representation and Behavior Tracing

arXiv preprint, Co-author

2026.05

- **Task:** Diagnose why vision-language-action models succeed or fail during embodied rollouts.
- **Method:** Collaborative work tracing representation dynamics and behavior signals across model layers, decisions, and rollout outcomes.
- **Result:** Connects internal VLA states with observable behavior for failure analysis beyond final task scores; no individual contribution claim is made beyond co-authorship.

Medical AI

VeloxSeg: JL-Guided Efficient 3D Medical Segmentation

ICLR 2026, First Author [CCF-A]

2026.04

- **Task:** Address volumetric complexity, multimodal complementarity, and deployment efficiency in PET/CT and MRI 3D segmentation.
- **Method:** Designed a lightweight CNN-Transformer for multi-scale cross-modal context, JL-guided local 3D features, and training-time texture-prior transfer.
- **Result:** Reached **62.51** / **56.48 Dice** on AutoPET-II / Hecktor2022 with **1.66M parameters**, improving GPU/CPU efficiency and memory.

H2ASeg: Hierarchical PET/CT Tumor Segmentation

MICCAI 2024, First Author [CCF-B]

2024.10

- **Task:** Frame 3D PET/CT tumor segmentation as multimodal volumetric perception combining PET metabolic localization with CT boundary cues.
- **Method:** Built hierarchical cross-modal fusion for local/global PET-CT interaction and target-aware feature reweighting.
- **Result:** Reached **60.03** / **59.69 Dice** on AutoPET-II / Hecktor2022; best-baseline gains: **+2.70 pp** / **+4.9 pp**.

⚙️ Core Methods & Engineering

Research: Video world models, generative-video evaluation, world-state persistence, embodied AI and VLA diagnostics; medical AI, PET/CT and MRI 3D segmentation

Engineering: Python, PyTorch, CUDA, DDP, Shell/Bash, $\text{\LaTeX}$; training/evaluation pipelines, experiment reproduction, ablation studies, custom model implementation

Tools: Linux, Git, Docker, Weights & Biases / TensorBoard, vision, multimodal, and medical-imaging data-processing toolchains

Writing: English manuscript writing for ICLR / MICCAI / Medical Image Analysis submissions, experiment analysis, and reproducibility documentation

📄 Full Research Outputs

<i>Video World Models & Generative Video</i>	2026 – Present
[1] Current World Models Lack a Persistent State Core, <i>First Author</i>	arXiv 2026

<i>Embodied AI & Representation Learning</i>	2025 – Present
[1] VLA-Trace: Diagnosing Vision-Language-Action Models through Representation and Behavior Tracing, <i>Co-author</i>	arXiv 2026
[2] Pelican-Unify 1.0: A Unified Embodied Intelligence Model, <i>Co-author</i>	arXiv 2026
[3] Does DINOv3 Set a New Medical Vision Standard?, <i>Co-first Author</i>	arXiv 2025

<i>Medical AI</i>	2023 – 2026
[1] Johnson-Lindenstrauss Lemma Guided Network for Efficient 3D Medical Segmentation, <i>First Author</i>	ICLR 2026 [CCF-A]
[2] H2ASeg: Hierarchical Adaptive Interaction and Weighting Network for Tumor Segmentation in PET/CT Images, <i>First Author</i>	MICCAI 2024 [CCF-B]
[3] AttriMIL: Revisiting Attention-based Multiple Instance Learning for WSI Classification, <i>Co-author</i>	Medical Image Analysis 2025 [CCF-C] [SCI Q1]
[4] IPGPhormer: Interpretable Pathology Graph-Transformer for Survival Analysis, <i>Co-author</i>	arXiv 2025
[5] A Localization-to-Segmentation Framework for Automatic Tumor Segmentation in Whole-Body PET/CT Images, <i>Co-author</i>	arXiv 2023

💡 Additional Project

Generative Sensor Signal Enhancement	Jun. 2023 – Jun. 2024
<i>Lead, National Undergraduate Innovation Program</i>	
Reproduced and compared GAN, diffusion, and flow-based models; designed conditional guidance strategies to improve reconstruction quality and perceptual fidelity, producing a reproducible comparison pipeline and experiment report.	